

DATA STRUCTURE AND ALGORITHM-1

PROJECT TOPIC – SMART RAILWAY MANAGEMENT SYSTEM

TEAM 1

ABSTRACT

The Smart Railway Management System is a software-based application designed to efficiently manage railway operations by integrating train administration, scheduling, and passenger ticket services into a single platform. The primary objective of this project is to automate core railway functions, reduce manual intervention, and improve accuracy, transparency, and user convenience.

The system includes a comprehensive **Train Management module for administrators**, allowing them to add new trains by entering details such as train number, train name, source, destination, total number of seats, and ticket price. These records are stored using structured data storage methods such as arrays of objects or data files. Administrators can also view, modify, or delete existing train details, ensuring that the system maintains up-to-date and accurate train information.

A dedicated **Schedule and Route Management module** enables administrators to set and update train schedules, including departure times, arrival times, and days of operation. This ensures effective planning and smooth coordination of railway services.

The system optionally supports **User Account Management**, allowing passengers to register and log in using basic authentication methods. This feature helps personalize services and securely manage booking records.

For passengers, the **Ticket Booking and Reservation module** provides essential functionalities such as searching for trains based on source, destination, and travel date. Efficient searching techniques like linear or binary search are applied depending on the data organization. Users can view seat availability in real time and proceed with ticket booking. During booking, the system allocates seats, updates availability instantly, and generates a unique Ticket ID or PNR. When seats are unavailable, a simple queue mechanism can be used to manage waiting lists.

The system also supports **Ticket Cancellation**, allowing users to cancel bookings using their PNR, automatically freeing seats and updating records. Additionally, passengers can view their booking history and check PNR status at any time.

Overall, the Smart Railway Management System demonstrates the effective use of data structures and algorithms to manage real-world railway operations. It enhances operational efficiency, improves user experience, and serves as a practical example of applying software engineering principles to transportation management systems.

TEAM MEMBERS CONTRIBUTION

- 1) AXIT KANDAWAL (CB.SC.U4AIE25005) – 1)Will be handling the codes of Train management (admin) function.2) Overall compilation of the project.
- 2) ABDUL JABBAR M (CB.SC.U4AIE25001) – 1)Will be handling the codes of user account management function.2) preparation of ppt
- 3) M.ASHWIN KUMAR (CB.SC.U4AIE25025) – 1)Will be handling the codes of Ticket booking and reservation(user) function. 2) Preparation of project report
- 4) RITESH KUMAR (CB.SC.U4AIE25043) – 1) Will be handling the codes of Schedule route management (Admin) function.2) preparation of abstract.